



Development of a multidimensional centrality metric for ranking nodes in complex networks

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ABSTRACT

Ranking nodes in complex networks is a fundamental task in network science, with significant applications in fields like social networks, bioinformatics, and information dissemination. Many existing methods struggle to efficiently capture the multifaceted relationships between nodes, particularly in large-scale networks. However, they often ignore the semantic relationships between nodes when gathering information. To address these challenges, this paper introduces SAASP, a Semi-local centrality measure that combines an Augmented graph with Average Shortest Path theory to efficiently identify influential nodes in complex networks. In SAASP, the augmented graph can represent global relationships between nodes as semantic similarities, allowing distant relationships to be considered in node ranking. Additionally, it incorporates an enhanced version of average shortest path theory to handle the computational complexity associated with large networks. By extracting local subgraphs for each node and considering the extended neighborhood concept, SAASP ensures scalability while preserving key structural properties. The integration of the augmented graph and average shortest path theory enables SAASP to accurately and efficiently identify influential nodes in large-scale complex networks with lower complexity. The effectiveness of the proposed metric is demonstrated through extensive experiments using the SIR (Susceptible-Infected-Removed) model and Kendall's coefficient, where it outperforms existing centrality measures on real-world datasets.

1. Introduction

The analysis of complex networks has become a critical area of research, encompassing various fields such as bioinformatics, social networks, and information dissemination [1,2]. Complex networks, characterized by intricate structures and interdependencies, pose significant challenges for researchers, particularly in identifying influential nodes [3]. The problem of identifying influential nodes in these networks is pivotal as it can influence the spread of information, the robustness of the network, and the overall dynamics within the system [1,4]. Influential nodes are those that, when removed or targeted, cause substantial changes in the network's structure or function [2]. Identifying these nodes is not only of theoretical interest but also has practical implications. For instance, in social networks, influential nodes can be targeted for effective marketing or information campaigns [5,6]. In biological networks, identifying key proteins or genes can aid in understanding diseases and developing treatments [7]. Similarly, in infrastructure networks, critical nodes can be monitored to prevent

cascading failures [8].

Centrality measures are widely used to identify influential nodes [9–11]. These measures quantify the importance of nodes within a network based on various measures. Centrality measures can be broadly categorized into three types: local, global, and semi-local [2,12]. Local centrality measures evaluate a node's importance based on its immediate neighbors [13]. Global centrality measures consider the node's role in the overall network structure, often involving computationally intensive processes [14]. Semi-local centrality measures strike a balance by incorporating information from both local and more distant nodes, offering a compromise between accuracy and computational efficiency [15]. Researchers have shown increasing interest in semi-local centrality measures due to their potential to provide more accurate node rankings without the high computational costs associated with global measures [1,2]. However, existing semi-local centrality measures face challenges such as ignoring global relationships and inefficiencies in large-scale networks [16,17]. This necessitates the development of more computationally efficient and accurate methods.

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The Average Shortest Path (ASP) theory [3] presents an attractive approach for enhancing centrality measures and identifying influential nodes. The Relative Change of ASP (RCASP) [1] is one such measure inspired by ASP theory. RCASP measures the change in the value of ASP in the network after the removal of a node and assigns a higher rank to the node with the highest rate of change. While RCASP provides a useful framework, it still has limitations. For example, RCASP tends to overlook global relationships by treating two non-adjacent nodes as unrelated, which is not always the case. Nodes that are distant in the network might still share semantic similarities. Additionally, extracting local subgraphs for ASP-based measures can be challenging in sparse networks. These concerns highlight the need for improved methods that consider both global relationships and local subgraph extraction.

Our study aims to address these gaps by introducing a semi-local centrality measure that combines an augmented graph with ASP theory. In this paper, we introduce SAASP (Semi-local Augmented Shortest Path), a novel centrality measure designed to efficiently identify influential nodes in complex networks. SAASP leverages the augmented graph to incorporate global semantic similarities and employs a semi-local structure to efficiently identify influential nodes. This combination allows SAASP to consider distant relationships and accurately rank nodes based on their impact on the network's structure. Semi-local graphs are extracted by considering the Extended Neighborhood (EN) concept. This concept defines different neighbor levels to identify related nodes. The measure facilitates distributed extraction of local subgraphs, ensuring computational efficiency and scalability for large-scale networks. By addressing the shortcomings of existing methods, SAASP offers a more accurate and computationally efficient solution for identifying influential nodes in complex networks.

The main contributions of this paper are summarized as follows:

- We propose a new centrality measure that combines an augmented graph with ASP theory to effectively identify influential nodes in complex networks.
- The augmented graph represents global semantic relationships, enabling the consideration of distant node relationships in the ranking process, thus addressing the limitations of traditional methods that overlook non-adjacent but related nodes.
- We redefine ASP theory in a semi-local structure, which enables distributed and efficient extraction of local subgraphs and identification of influential nodes with reduced computational complexity.

The paper is organized as follows. Section 2 provides a brief overview of centrality measures and the existing literature. In Section 3, the proposed methodology for identifying influential nodes is explained. Numerical simulations are presented in Section 4 to demonstrate the effectiveness of the proposed centrality measure. Finally, some conclusions are presented in Section 5.

2. Related works

Identifying influential nodes in complex networks has been a focal point of research in various domains, including social networks, biological networks, and information dissemination [18,19]. Traditional centrality measures, such as Degree Centrality (DC) [18], Betweenness Centrality (BC) [19], and Closeness Centrality (CC) [20], have been widely used to rank nodes based on their importance. However, these methods often fall short in capturing the full spectrum of node influence, particularly in large-scale and sparse networks. For instance, degree centrality merely counts the number of direct connections a node has, neglecting the broader network structure [21,22]. Betweenness centrality, while accounting for the number of shortest paths passing through a node, becomes computationally expensive for large networks [23–25]. Closeness centrality, which measures the ASP from a node to all other nodes, similarly struggles with computational efficiency and may not reflect true influence in networks with varying topologies [26].

These limitations underscore the need for more sophisticated approaches that can consider both local and global network properties without excessive computational demands [27,28].

In response to these challenges, recent research has explored semi-local centrality measures and the incorporation of semantic relationships [29,30]. Semi-local methods, which balance between local and global network information, offer a promising avenue for more accurately identifying influential nodes. For example, the use of k-shell [31] decomposition provides insights into the core structure of networks but may still overlook semantic and distant relationships [32–34]. Moreover, many existing approaches do not scale well with network size, making them impractical for real-world applications where networks can be vast and dynamically evolving [35,36]. Recent advancements in the literature have introduced more sophisticated methods for identifying influential nodes in complex networks, aiming to address the shortcomings of traditional centrality measures [37–39]. One such approach is the Collective Influence (CI) [40], which extends the concept of node influence by considering the collective effect of nodes within a certain radius, enhancing its effectiveness in large-scale networks [40]. Another advanced method is the PageRank [41], originally developed for ranking web pages, which evaluates node importance based on the structure of incoming links, thereby providing a more nuanced measure of influence. Additionally, methods incorporating community structure, such as Community-Aware Centrality (CAC) [42], have been proposed to improve influence detection by leveraging the modular organization of networks. These methods focus on both intra-community and inter-community connections, offering a more comprehensive view of node influence. Furthermore, techniques utilizing machine learning, such as those based on Graph Neural Networks (GNNs), have gained traction [43]. GNNs can learn complex patterns in network data, providing a powerful tool for predicting node importance by capturing intricate dependencies between nodes [44,45]. These advanced methodologies reflect the ongoing efforts to develop more accurate and scalable solutions for identifying influential nodes in complex networks.

In this study, the aim is to identify the influential nodes in a complex network $G = (V, E)$, where $V = \{v_1, \dots, v_n\}$ is the set of nodes and $E = \{e_1, \dots, e_m\}$ represents the set of edges. According to the set E , let $A = (a_{ij})_{n \times n}$ be an adjacency matrix where $a_{ij} = 0$ indicates the existence of an edge between v_i and v_j , while $a_{ij} = 1$ indicates the absence of an edge between these nodes. The number of edges connected to each node is defined as the degree of that node. For example, $k_i = \sum_{v_j \in V} a_{ij}$ is defined as the degree of node v_i . Also, the set of neighbors associated with node v_i is defined by $\Gamma(v_i)$. Meanwhile, $\Gamma_l(v_i)$ is the set of neighbors associated with v_i with maximum hop equal to l . According to these definitions, Table 1 provides details of some traditional centrality measures.

Advanced semi-local centrality measures are of significant importance in the analysis of complex networks because they effectively identify influential nodes by leveraging both local and semi-local information [46]. Unlike local measures that only consider immediate neighbors, and global measures that require extensive computations, semi-local measures balance accuracy and computational efficiency by incorporating information about distant neighbors and semi-local structures [47,48]. This unique capability makes semi-local centrality measures highly effective and efficient for various applications, such as identifying critical nodes in social networks, optimizing distribution networks, and analyzing biological networks [49]. Next, some advanced semi-local centrality measures are reviewed.

Shao et al. [50] proposed the NL (Node Local) centrality measure, which considers the influence of a node's neighbors within 2-steps and the importance of its edges. This semi-local centrality measure evaluates the connectivity and unsubstitutability of edges to determine the node's topological position. By incorporating the impact of second-order neighbors and edge characteristics, NL centrality effectively measures node importance using both semi-local structure and topological

Table 1

An overview of traditional centrality measures

Measures	Formula	Parameters	Category	Complexity
DC [29]	$C_D(v_i) = k_i$	—	Local	$O(n)$
BC [30]	$C_B(v_i) = \sum_{v_j \neq v_i, v_j \in V} \frac{\sigma_{j,k}(v_i)}{\sigma_{j,k}}$	$\sigma_{j,k}(v_i)$: Shortest paths through v_i ; $\sigma_{j,k}$: Total shortest paths between v_j and v_k	Global	$O(n^3)$ (using Floyd-Warshall)
CC [31]	$C_C(v_i) = \frac{1}{\sum_{v_j \neq v_i, v_j \in V} d(v_i, v_j)}$	$d(v_i, v_j)$: Shortest path distance between v_i and v_j	Global	$O(n^3)$ (using Floyd-Warshall)
k-shell [37]	$C_{KS}(v_i) = k - \text{shell index of } v_i$	—	Semi-Local	$O(n + m)$
CI [40]	$C_{CI}(v_i) = (k_i - 1) \sum_{v_j \in \text{Ball}(v_i, r)} (k_j - 1)$	r : Measuring radius; $\text{Ball}(v_i, r)$: Nodes within radius r of v_i	Semi-Local	$O(lm)$
PageRank [41]	$C_{PR}(v_i) = \frac{1 - \rho}{n} + \rho \sum_{v_j \in \Gamma(v_i)} \frac{C_{PR}(v_j)}{k_j}$	ρ : Damping factor; $\Gamma(v_i)$: Set of nodes linking to v_i	Global	$O(m)$ per iteration
CAC [42]	$C_{CAC}(v_i) = \alpha \frac{k_i^{\text{intra}}}{K^{\text{intra}}} + (1 - \alpha) \frac{k_i^{\text{inter}}}{K^{\text{inter}}}$	α : Weighting factor; k_i^{intra} : Intra-community links; K^{intra} : Total intra-community links; k_i^{inter} : Inter-community links; K^{inter} : Total inter-community links	Semi-Local	$O(n + m)$

positioning. Hajarathaiah et al. [51] introduced the Local Relative ASP (LRASP) centrality measure based on a node's impact on the local network's ASP when removed. The LRASP centrality offers efficient computation time and superior performance in identifying influential nodes compared to other centrality measures like degree, betweenness, closeness, semi-local, PageRank, Trust-PageRank, and global LRASP, thereby enhancing information diffusion analysis in networks.

Xiao et al. [52] introduced a new semi-local centrality measure called LASPN (Local ASP with extended Neighborhood concept) to identify influential nodes in complex networks. LASPN extracts subgraphs associated with each node and applies the ASP theory to these subgraphs. Utilizing an extended neighborhood concept, LASPN efficiently identifies the nearest neighbors of each node, making it suitable for large-scale networks. The measure ranks nodes by considering both the relative changes in the ASP and the importance of the nodes and their nearest neighbors. Zhang et al. [53] proposed the Semi-LRASP (SLRASP) centrality measure for ranking influential nodes, which builds on the LRASP measure. This approach involves extracting semi-local subgraphs from fixed-length neighborhoods to evaluate nodes. The SLRASP measure offers a global importance model for assessing the overall influence of nodes by analyzing the relative change in the ASP across the network. Table 2 provides details of some advanced centrality measures.

3. Methodology

The proposed SAASP centrality measure is designed to address the limitations of traditional centrality measures by balancing accuracy and computational efficiency. Unlike local centrality measures, which rely only on direct neighbors, or global centrality measures, which consider the entire network, SAASP incorporates information from a semi-local perspective by evaluating nodes within a defined neighborhood radius. SAASP begins by constructing local subgraphs for each node,

capturing multi-hop neighborhood information to provide a more comprehensive understanding of each node's influence. These local subgraphs are then embedded into an augmented graph, which enhances the original network by incorporating additional relational information through refine, auxiliary, and enhance modules. Finally, the ASP theory is employed to evaluate the average distance from each node to others within this augmented graph, allowing SAASP to effectively identify nodes that are both locally and globally central. In addition to ASP theory, SAASP considers the importance of the node itself and its neighbors to calculate the centrality. Since applying ASP theory to the entire network is not cost-effective in terms of computational complexity, we present a distributed approach based on the EN concept to extract local subgraphs for each node. Generally, SAASP calculates the influence of each node by combining four components: influence of the node itself, influence of the node's neighbors, influence based on ASP theory, and influence based on semantic similarity. We calculate the influence score for undirected and unweighted graphs. Fig. 1 shows the proposed SAASP architecture for node ranking.

SAASP introduces an augmented graph that enhances the representation of the network by integrating global, local, and feature-based similarities between nodes. This approach allows SAASP to capture both immediate and distant relationships within the network, making it more robust, particularly in sparse networks or those with complex connectivity patterns. The augmented graph in SAASP measure operates in three key stages: Refined Graph, Auxiliary Graph, and Enhanced Graph. The integration of refined, auxiliary, and enhanced graphs within the SAASP framework results in a more accurate and computationally feasible method for identifying influential nodes in complex networks. Meanwhile, the ASP theory in SAASP framework plays a critical role in accurately capturing the influence of nodes by considering the shortest paths within the augmented network. ASP theory measures the average distance from a given node to all other nodes within the network, providing insight into how efficiently information

Table 2

An overview of advanced centrality measures

Measures	Formula	Parameters	Complexity
NL [50]	$C_{NL}(v_i) = \frac{1}{L \cdot n} \left(\alpha \cdot k_i + \beta \cdot \sum_{l=1}^L \sum_{v_j \in V^{+l}(v_i)} \frac{k_j}{l} \right)$	L : Neighborhood length; $V^{+l}(v_i)$: set of nodes with maximum l -hop from v_i ; α, β : Impact coefficients	$O(L \cdot n + m)$
LRASP [51]	$C_{LRASP}(v_i) = \frac{ \text{ASP}[\widehat{G}_L(v_i)] - \text{ASP}[G_L(v_i)] }{\text{ASP}[G_L(v_i)]}$	$\text{ASP}[G]$: Average length of the shortest paths in the network; $\widehat{G}_L(v_i)$: Subgraph $G_L(v_i)$ after removing v_i	$O(L \cdot n \cdot \log n + m)$
LASPN [52]	$C_{LASPN}(v_i) = \xi \frac{k_i}{n} + \delta \cdot \left[\frac{1}{ G_L(v_i) } \sum_{l=1}^L \sum_{w \in \Gamma_l(v_i)} \sqrt{\frac{\widehat{w}_{u,v} \cdot k_l}{d(v_i, v_j) \cdot (k_i + k_j)}} \right] + \gamma \cdot C_{LRASP}(v_i)$	$G_L(v_i)$: Local subgraph for v_i based on L ; ξ, δ, γ : Impact coefficients	$O(L \cdot n + m \cdot \log n + n)$
SLRASP [53]	$C_{SLRASP}(v_i) = \frac{\alpha \cdot k_i + \beta \cdot \sum_{l=1}^L \sum_{v_j \in V^{+l}(v_i)} \frac{k_j}{l}}{L \cdot N_L(v_i)} + \gamma \cdot C_{LRASP}(v_i)$	$N_L(v_i)$: Size of the extracted subgraph for v_i through L ; α, β, γ : Impact coefficients	$O(L \cdot n \cdot \log n + m)$

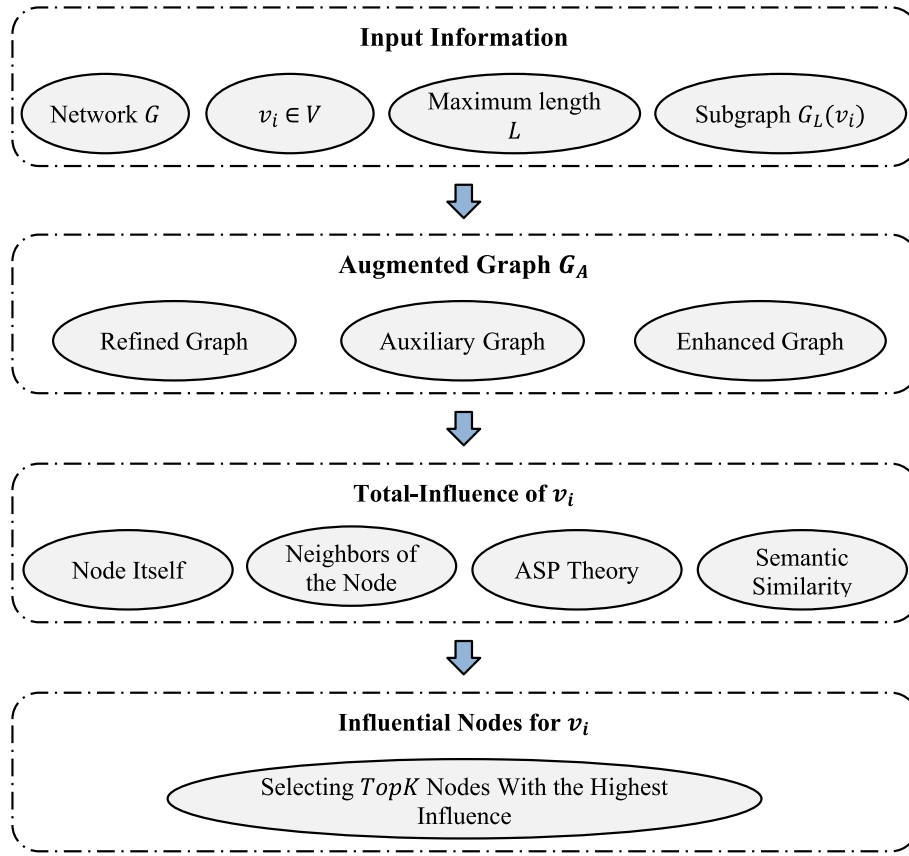


Fig. 1. Proposed SAASP architecture for node ranking.

or influence can spread from that node. By integrating ASP into SAASP, the measure not only evaluates the proximity of a node to others but also considers the enhanced connectivity and relationships introduced through the augmented graph.

3.1. Local subgraphs

In centrality measures, each node in a network is assigned an influence score, which is then used to rank the nodes based on their impact within the network. Local centrality measures calculate a node's influence score using only information from directly connected neighbors, whereas global centrality measures consider information from the entire network. Naturally, the simplicity and low computational complexity of local centrality measures come at the cost of lower accuracy, while global centrality measures offer higher accuracy but at the expense of increased computational complexity. Semi-local centrality measures aim to strike a balance between accuracy and complexity by considering information from neighboring nodes within a limited path length.

A key challenge in semi-local centrality measures is the identification of neighboring nodes within the specified path length. Identifying 1-hop neighbors can be done through the adjacency matrix; however, identifying L -hop neighbors (for $L > 1$) is more challenging, especially when dealing with large-scale networks. To address this challenge, the proposed SAASP leverages the EN concept and a distributed approach to identify neighboring nodes up to a maximum path length of L . SAASP can create a neighborhood table for each node by exchanging control messages based on the parameter L , at specific time intervals between nodes. Using this neighborhood table, a local subgraph comprising neighbors with a maximum path length of L can be generated for each node. This means that each node can independently create a local subgraph by communicating with its adjacent nodes.

Let $G_L(v_i)$ represent a local subgraph extracted from G for node v_i

with a maximum path length of L . The paths between v_i and other nodes in $v_j \in G_L(v_i)$ include 1-hop, 2-hop, ..., up to L -hop paths. Inspired by methods presented in [1,2], each node creates a neighborhood table by sending control messages "P1" and "P2" at specific time intervals. Fig. 2 illustrates the overall structure of control message exchanges used to create the neighborhood table. Here, v_i , as the target node, broadcasts the control message "P1" with "PathLength = l " across the network. At first l is initialized with L . This message is sent from v_i to all nodes in $v_j \in G_L(v_i)$. Each node that receives this message forwards it again with "PathLength = $l - 1$ " until "PathLength = 1" is reached. Meanwhile, every node that receives "P1" sends the control message "P2" back to the target node along the recorded path. In this message, each node appends its identifier along with the path length. The target node checks the received messages and removes duplicate information. The neighborhood table at the target node is updated with the control messages "P2" received at different time intervals. Using this method, SAASP can extract a local subgraph for each node in a distributed manner.

3.2. Augmented graph

Most existing centrality measures focus on calculating the similarity between two directly connected nodes. Meanwhile, some semi-local centrality measures calculate the similarity between two nodes that are connected by at least one path of finite length. However, any pair of nodes in the network may have a semantic similarity, even if they are very far apart. This issue becomes more pronounced in sparse networks, where many nodes are connected by long paths. Also, considering the equal contribution of all nodes in the calculation of similarity regardless of the semantics of each node has a significant effect on the ranking of nodes. Specifically, semantically related connections between certain nodes may have a higher influence on their impact compared to others. Semantic similarity in a directed or undirected graph involves the

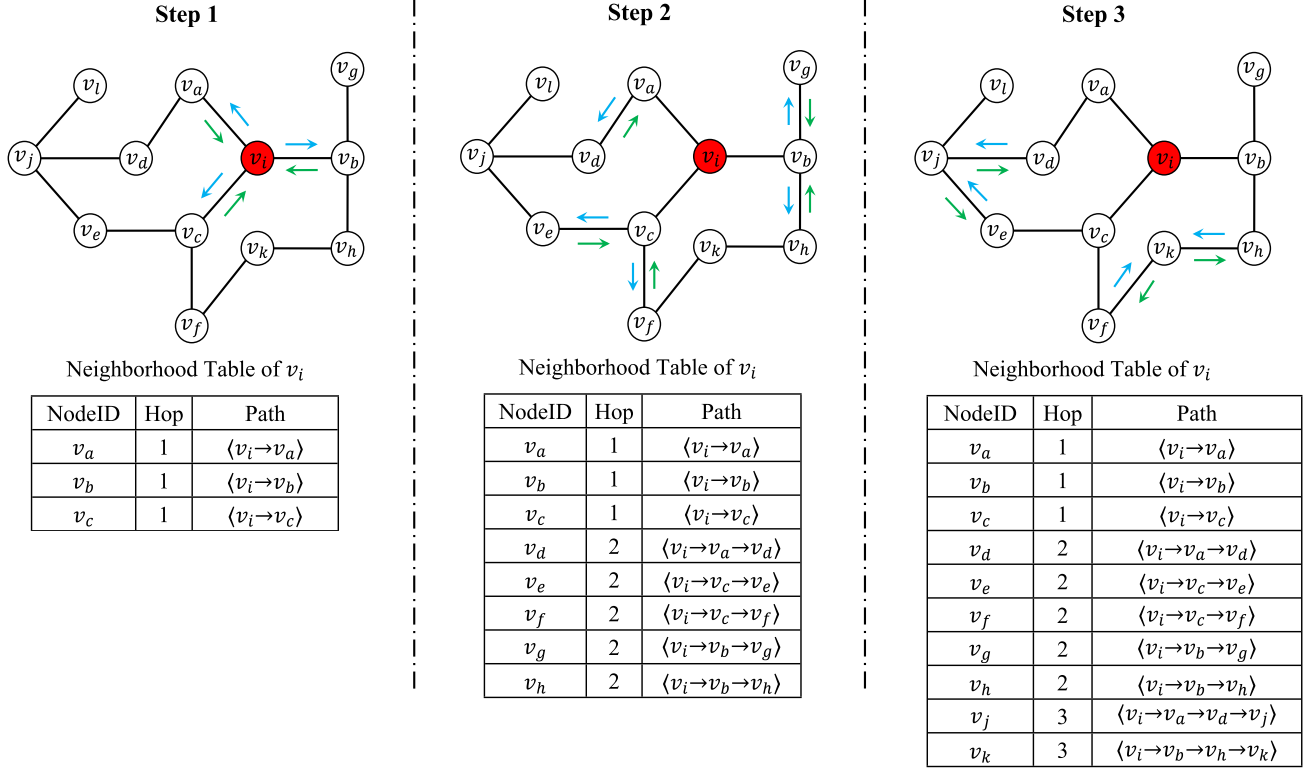


Fig. 2. Overview of extracting local subgraphs based on control messages “P1” and “P2”.

semantic relationships between nodes. To more accurately calculate the similarity between nodes, SAASP creates an augmented graph that includes refined, auxiliary and enhanced graphs. Augmented graphs use both structure and feature spaces to improve information representation. This graph deals with the two problems of similarity between distant nodes and sparse networks by considering semantic relationships. Therefore, considering semantic relationships can play an important role in ranking nodes.

Each node in the network is associated with different semantics. Here, each node contains a set of interests described by a feature vector. Since we are dealing with undirected and unweighted networks, the feature vector for each node is generated based on the three factors of the node's own degree, the average degree of its neighbors, and the maximum degree of its neighbors. The adjacency matrix A is defined as a node-node matrix that contains close relationships between each pair of connected nodes. Meanwhile, augmented graph considering both structure and feature space is defined by matrix G_A which can represent the semantic similarity between nodes. As shown in Fig. 3, the G_A matrix is a combination of the refined, auxiliary, and enhanced graphs.

Let the weight matrix W_R represent the refined graph that can strengthen the adjacency matrix A by improving the network repre-

sentation. By recording the global relations, W_R can achieve a good level of performance compared to the adjacency matrix A . By joining the adjacency matrix A and the identity matrix I_n , W_R is defined as follows:

$$W_R = (k_{avg})^{-1} \cdot (I_n + A). \quad (1)$$

where k_{avg} is the average degree of the entire network. By multiplying the average degree in matrix $(I_n + A)$, an $n \times n$ matrix is obtained.

Let the weight matrix W_U represent the auxiliary graph that uses the similarity matrix $\mathcal{W}$ to enhance the adjacency matrix A . The matrix $\mathcal{W}$ is calculated based on the feature matrix $X \in \mathbb{R}^{n \times d}$ by $w_{ij} = \text{COS}(x_i, x_j^T)$ for each pair of nodes v_i and v_j . Here, n is the number of nodes, d is the number of features for each node, x_i is the feature vector for node v_i , and COS is the cosine similarity function. The feature vector for each node is created by the three factors of the node's own degree, the average degree of its neighbors, and the maximum degree of its neighbors. Since the matrix $\mathcal{W}$ is built on the basis of the feature matrix X , it contains significant noise. Hence, the k highest values of this matrix are kept to reduce the noise in $\mathcal{W}$. Let $\mathcal{W}_k = \text{TOP}(\text{Descending_Order}(\mathcal{W}), k)$ be a function to select the k highest values from $\mathcal{W}$. Specifically, $\mathcal{W}_k$ and $w_i^T(k)$ are formulated as follows:

$$\mathcal{W}_k = [w_1(k), w_2(k), \dots, w_n(k)] \in \mathbb{R}^{n \times n}, \quad (2)$$

$$w_i^T(k) = [w_{i,1}(k), w_{i,2}(k), \dots, w_{i,n}(k)] \in \mathbb{R}^{1 \times n}, \quad (3)$$

The $\mathcal{W}_k$ matrix is normalized to better represent the similarity between nodes:

$$\mathcal{W}_k = \frac{\mathcal{W}_k}{\max(\mathcal{W}_k)}, \quad (4)$$

Since $\mathcal{W}_k$ is constructed based on the feature matrix X , it may have significant computational complexity for large graphs. Therefore, we assume that this matrix is computed offline and is available prior to any calculations, as we consider the network to be static and the features do

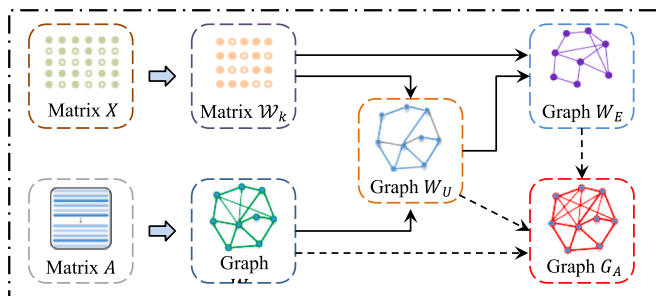


Fig. 3. Augmented graph architecture.

not change over time.

The weighted matrix W_U , which represents the auxiliary graph, utilizes the matrix W_R to provide a more accurate representation of information compared to the adjacency matrix A . Additionally, W_U includes the semantic similarity between nodes, as it considers global relationships between nodes through $\mathcal{W}_k$. Accordingly, the weighted matrix W_U is constructed based on the matrices $\mathcal{W}_k$ and W_R as follows:

$$W_U = \alpha \cdot \mathcal{W}_k + (1 - \alpha) \cdot W_R, \quad (5)$$

where α is a constant coefficient to consider the effect of $\mathcal{W}_k$ relative to W_R in the calculation of W_U . Empirically, we found a suitable value for α to be 0.6.

Let the weight matrix W_E represent the enhanced graph that is created through the heterophilic graph. Let the weight matrix W_E represent the enhanced graph that is created through the heterophilic graph. Heterophilic graphs are a type of graph where nodes that are connected tend to have different characteristics or attributes. In these graphs, there is a higher likelihood of connections between nodes with dissimilar properties or that belong to different categories. This is in contrast to homophilic graphs, where nodes with similar characteristics are more likely to connect. Considering [100], heterophilic graphs can model nodes with long distances using structural and feature information. According to this, the weight matrix W_E is defined as a heterophilic graph by combining the matrices W_U and $\mathcal{W}_k$:

$$W_E = (\hat{1} - \mathcal{W}_k) \odot (\hat{1} - W_U), \quad (6)$$

where $\hat{1}$ is an $n \times n$ matrix with all values equal to 1. Also, the symbol $\odot$ represents the Hadamard product. The Hadamard product is a binary operation that takes two matrices of the same dimensions and produces another matrix of the same dimensions. The elements of the resulting matrix are the products of the corresponding elements from the input matrices. For two matrices $\mathcal{A}$ and $\mathcal{B}$ of the same size $m \times n$, the Hadamard product $\mathcal{C} = \mathcal{A} \odot \mathcal{B}$ is defined as $c_{ij} = a_{ij} \cdot b_{ij}$. This means that each element $c_{ij} \in \mathcal{C}$ in the resulting matrix is simply the product of $a_{ij} \in \mathcal{A}$ and $b_{ij} \in \mathcal{B}$, where a_{ij} and b_{ij} are the elements in the same position in the original matrices. So, this operation is different from matrix multiplication, which involves summing the products of rows and columns. The Hadamard product is purely element-wise.

Eventually, the augmented graph G_A is formed by integrating the refined graph W_R , auxiliary graph W_U , and enhanced graph W_E :

$$G_A = \xi_1 \cdot W_R + \xi_2 \cdot W_U + \xi_3 \cdot W_E. \quad (7)$$

where ξ represents the coefficients associated with the three matrices W_R , W_U , and W_E , which are empirically set to 0.33 to provide optimal performance. To reduce the effect of noise, $G_A =$

$TOP(Descending_Order(G_A), k)$ is applied to keep only the k highest values.

For more clarity, an example of the process of creating an augmented graph with $k = 4$ is presented in Fig. 4.

3.3. ASP theory

The ASP theory is a concept in graph theory and network analysis that refers to the average length of the shortest paths between all pairs of nodes in a graph [1]. It is a measure of the efficiency of information or signal transmission within a network. The ASP is important in understanding the overall connectivity and navigability of the network. In a graph, the shortest path between two nodes is the path that minimizes the total distance or the number of edges traversed between those nodes. The distance is often defined by the number of edges in unweighted graphs or the sum of edge weights in weighted graphs. The ASP is calculated by finding the shortest path between all possible pairs of nodes in the graph and then averaging these path lengths. Mathematically, if $d(v_i, v_j)$ represents the shortest path distance between node v_i and node v_j , and n is the total number of nodes, the ASP is given by:

$$ASP[G] = \frac{1}{n(n-1)} \sum_{v_i \neq v_j \in G} d(v_i, v_j), \quad (8)$$

The ASP provides a single number that summarizes the efficiency of communication within the network. A smaller ASP indicates that, on average, any two nodes in the network can communicate through a relatively short path, suggesting high efficiency and connectivity. Conversely, a larger ASP suggests that communication between nodes is less efficient, potentially indicating a more dispersed or less connected network structure.

RCASP is a measure used in network analysis to measure the impact of specific changes or interventions in a network on its overall connectivity. Specifically, RCASP quantifies how the ASP length in a network change relative to its original state after certain modifications, such as the addition or removal of nodes or edges. The RCASP focuses on the relative difference in the ASP before and after a change in the network. Consider a social network where we analyze the impact of removing a highly connected individual (node). The removal might increase the ASP if the individual served as a key bridge between communities. The RCASP would capture the relative increase in the ASP, reflecting how much the network's overall communication efficiency has declined due to this change. RCASP is calculated as the ratio of the change in ASP to the original ASP:

$$RCASP[\tilde{G}] = \frac{|ASP[G] - ASP[\tilde{G}]|}{ASP[G]}. \quad (9)$$

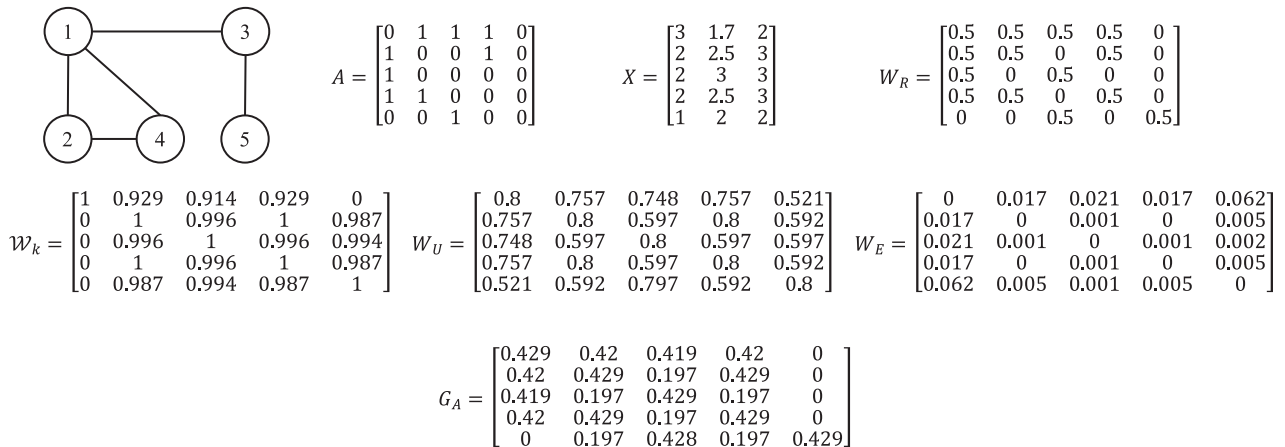


Fig. 4. An example of the process of creating an augmented graph with $k = 4$.

where $ASP[G]$ is the ASP length before the change, and $ASP[\tilde{G}]$ is the ASP length after the change.

3.4. SAASP centrality measure

The proposed centrality measure, SAASP, calculates the influence of each node by integrating four key components. First, it assesses the influence of the node itself, considering its inherent properties and direct connections. Second, SAASP evaluates the influence of the node's neighbors, extending the analysis to include multi-hop connections and providing a deeper understanding of the node's relative importance within its local network context. Third, it incorporates the ASP theory, measuring the node's efficiency in connecting with other nodes and reflecting its global influence. Finally, SAASP accounts for semantic similarity, considering the relationships between nodes that may not be immediately apparent through direct connections but are significant due to shared characteristics. By combining these components, SAASP provides a comprehensive and nuanced assessment of node influence, capturing both local and global network dynamics and recognizing meaningful indirect relationships.

3.4.1. Influence of the node itself

The importance of the node itself as local-influence is calculated based on the degree ratio of the node. Local-influence for each $v_i \in V$ is calculated as follows:

$$C_{LI}(v_i) = \frac{k_i}{k_{max}}, \quad (10)$$

where k_{max} is the maximum degree of nodes.

3.4.2. Influence of the node's neighbors

The importance of the neighbors of the node as semi-local-influence is calculated based on the three attributes of distance, weight, and degree by considering a damping coefficient. Semi-local-influence for each $v_i \in V$ is calculated as follows:

$$C_{SLI}(v_i) = \sum_{l=1}^L \left(\lambda^l \cdot \sum_{v_j \in I_l(v_i)} \frac{w_{ij} \cdot k_j}{d(v_i, v_j)} \right), \quad (11)$$

where λ is a damping factor for applying higher importance to neighbors with shorter distances.

3.4.3. Influence based on ASP theory

This component is defined as ASP-influence. We use the RCASP measure, which is inspired by ASP theory and EN concept, to calculate this component. Since the calculation of ASP on the whole graph G is computationally complex, we calculate the ASP on the extracted local subgraphs for each node. As a reminder, $G_L(v_i)$ is the local subgraph for node v_i which is extracted based on the concept of EN. Meanwhile, let $\tilde{G}_L(v_i)$ be the subgraph of $G_L(v_i)$ after removing node v_i . Obviously, $RCASP[G_L(v_i)]$ calculates the relative changes made in the structure of the graph $G_L(v_i)$ after removing node v_i . The RCASP criterion is defined by considering $G_L(v_i)$ and $\tilde{G}_L(v_i)$ as follows:

$$C_{ASPI}(v_i) = RCASP[G_L(v_i)] = \frac{|ASP[G_L(v_i)] - ASP[\tilde{G}_L(v_i)]|}{ASP[G_L(v_i)]}, \quad (12)$$

3.4.4. Influence based on semantic similarity

SAASP also considers the semantic similarity between nodes, identifying and leveraging relationships that may not be immediately apparent through direct connections. This component allows SAASP to account for nodes that, despite being far apart in the network, share significant similarities and thus have a meaningful influence on each

other. Let this component be calculated as semantic-influence as follows:

$$C_{SI}(v_i) = \sum_{v_j \in G_A} G_A(v_i, v_j), \quad (13)$$

where $G_A(v_i, v_j)$ is the element associated with nodes v_i and v_j of the augmented graph G_A .

Considering the four defined components, the total-influence of v_i node based on SAASP is calculated as follows:

$$C_{SAASP}(v_i) = \xi_1 \cdot C_{LI}(v_i) + \xi_2 \cdot C_{SLI}(v_i) + \xi_3 \cdot C_{ASPI}(v_i) + \xi_4 \cdot C_{SI}(v_i). \quad (14)$$

where ξ_1, ξ_2, ξ_3 , and ξ_4 are influence coefficients, each empirically set to 0.25 for optimal performance.

4. Experimental simulations

This section is related to the validation of SAASP in comparison to traditional centrality metrics and advanced methods. We employ several real-world network topologies of various scales for simulation purposes. All implementations are conducted using MATLAB R2021a on Intel Core i7 processors with 32GB of RAM. Influential nodes across all benchmark methods are identified under the same settings. In each centrality metric, the top-ranked nodes, $TopK$, are selected as seed nodes. In line with existing works, we set $TopK = 10$ as the default value in our simulations [52,53]. To evaluate and compare the results of different methods, we use Kendall's τ coefficient [54,55], which measures the ordinal association between the rankings produced by different centrality metrics. This approach allows us to assess the consistency and performance of SAASP relative to the baseline methods across diverse network structures and topologies.

4.1. Datasets description

To evaluate the proposed SAASP, extensive experiments were conducted on six real-world datasets, including both homophilic and semantic networks. All homophilic datasets consist of unweighted and undirected graphs that were selected at various scales. Meanwhile, the semantic datasets contain a vector of semantic features for each node, capturing the underlying relationships between them. The statistical information for these datasets is summarized in Table 3. These datasets provide a diverse testing environment, allowing us to assess the performance of SAASP across different types of networks, including those with structural similarities and those enriched with semantic information.

We use six real-world datasets for our simulations: Email, Yeast, Router, Twitter, Rice-Facebook, and DBLP. The Email dataset represents communication between individuals in an email network, often used to study information flow and social connections. The Yeast dataset is a well-known biological network used in various studies, particularly in the fields of systems biology and bioinformatics. It represents interactions between proteins in the yeast *Saccharomyces cerevisiae*, focusing on the relationships that define various cellular processes. The Router dataset models the internet's underlying structure, where routers are connected by physical links, highlighting network robustness. The Twitter dataset focuses on social interactions through follower relationships, with nodes as users and edges as followerships, often employed to study social influence and information dissemination. The Rice-Facebook dataset represents the social network of students at Rice University, where nodes are students and edges represent Facebook friendships, offering insights into social group dynamics. Finally, the DBLP dataset is a co-authorship network in computer science, where nodes are authors and edges indicate collaborations, making it useful for studying academic collaboration and influence.

Table 3

The statistics of six real-world datasets

Type	Dataset	#Nodes	#Edges	#Attributes	k_{avg}	k_{max}
Homophilic	Email	1133	5451	–	9.62	71
	Yeast	2361	7182	–	6.08	101
	Router	5022	6258	–	2.49	106
Semantic	Twitter	3560	6677	3	3.56	36
	Rice-Facebook	1205	42,443	2	7.44	45
	DBLP	317,080	1,049,866	4	6.62	429

4.2. Evaluation criteria

The performance of centrality criteria is evaluated and compared based on the spread of influence and running time. In this study, the Susceptible-Infected-Recovered (SIR) model [49] is applied as an epidemic model to simulate centrality measures on real-world network topologies to compare the performance of SAASP with other existing approaches. Here, Kendall's τ coefficient is used to check the agreement of the actual ratings with the ratings obtained by the SIR model.

The SIR model, a well-known epidemic model, categorizes nodes into three states: susceptible (S), infected (I), and recovered (R) [49]. Nodes transition from susceptible to infected upon contact with infected nodes, and later recover after a certain period [56,57]. The model's dynamics are governed by transmission (λ) and recovery (μ) rates, which determine how quickly the infection spreads and recovers. The performance of centrality criteria is evaluated and compared based on the spread of influence and running time. In this study, the SIR model is applied as an epidemic model to simulate centrality measures on real-world network topologies to compare the performance of SAASP with other existing approaches. The SIR model operates under the premise that once individuals enter the recovered group, they gain permanent immunity to the disease. The rumor propagation dynamics of the SIR model are represented by Eqs. (15) and (16). By applying the SIR model to various centrality measures, the study evaluates the influence each node has on the network, allowing a direct comparison with SAASP.

$$S(i) + I(j) \xrightarrow{\lambda} I(i) + I(j), \quad (15)$$

$$I(i) \xrightarrow{\mu} R(i), \quad (16)$$

where i and j denote two nodes within the network, while λ and μ correspond to the infection rate and recovery rate, respectively.

To quantify the correlation between the influence rankings generated by SAASP and those obtained from the SIR simulations, Kendall's τ coefficient is used. Kendall's τ is a statistic that measures the agreement between two ranked lists by comparing pairs of items. For each pair of nodes, the rankings are considered concordant if the relative order is the same in both lists, and discordant if the order is reversed. Kendall's τ coefficient is defined as:

$$\tau = \frac{n_c - n_d}{0.5n(n-1)}, \quad (17)$$

where n_c and n_d are the numbers of concordant and discordant pairs, respectively, and n is the total number of nodes. A value of τ close to 1 indicates strong agreement between the rankings, while a value close to -1 indicates strong disagreement.

By comparing the τ values for different centrality measures, the effectiveness of SAASP in identifying influential nodes is validated. In the calculation of Kendall's τ coefficient, the function $F(t)$ is used to determine the concordance and discordance between pairs of rankings. Specifically, a pair of observations (x_i, y_i) and (x_j, y_j) is considered concordant if the ranks for both observations are in the same order. That is, if $x_i < x_j$ and $y_i < y_j$ or if $x_i > x_j$ and $y_i > y_j$. Meanwhile, a pair is discordant if the ranks for the observations are in different orders, meaning if $x_i < x_j$ and $y_i > y_j$ or if $x_i > x_j$ and $y_i < y_j$. By using $F(t)$,

Kendall's τ provides a measure of the strength and direction of association between the two rankings. The function $F(t)$ counts the number of concordant and discordant pairs in the ranked lists at a specific time t :

$$F(t) = n_{I(t)} + n_{R(t)}. \quad (18)$$

where $n_{I(t)}$ and $n_{R(t)}$ are the number of infected and recovery nodes at time t .

4.3. Benchmark methods

In this study, the proposed SAASP centrality measure is compared with several classic centrality measures, including DC [29], BC [30], CC [31], and CI [40]. Additionally, SAASP is evaluated against SLRASP [53] as an equivalent and advanced approach. All these metrics are simulated using MATLAB under identical settings to ensure a fair comparison. Furthermore, the results are reported based on the average of 25 runs to provide reliable comparisons.

4.4. Results and analysis

The results related to the evaluation of SAASP in comparison with classic and advanced centrality measures are presented in this section. The experiments were conducted with parameters set at $\lambda = 0.1$ and $\mu = 1$, while the default value for the neighborhood level parameter, L , is set to 3 [2]. The findings confirm the superiority of our metric in identifying influential nodes. In our initial experiment, we analyze the weighted score computed for SAASP. Table 4 illustrates the average total influence for the ten identified seed nodes across each dataset. This analysis evaluates SAASP against SLRASP, as both methods are semi-local and share similar architectures. The results are reported for neighborhood levels 2, 3, and 4. Across all datasets, the proposed SAASP achieves optimal results based on features derived from semantic similarity and ASP theory. By examining the centrality scores, it becomes evident that SAASP provides slightly higher scores compared to SLRASP. Empirical tests conducted on six datasets demonstrate that incorporating semantic similarity and ASP theory to define centrality enables our metric to obtain more comprehensive information about the nodes, thereby enhancing the reliability of the empirical results.

In this study, the SIR model, widely applied in the context of epidemic spread in networks, is utilized to measure the influence of nodes by analyzing their propagation patterns and differences across various nodes. The SIR model identifies which nodes can disseminate information or infection more quickly and extensively over time. Typically, a node's significance is directly related to its ability to propagate. The node with the highest spreading potential holds the greatest influence. Tables 5 and 6 presents the top-10 nodes from each dataset, ranked by their spreading capabilities according to DC, BC, CC, CI, SLRASP, and SAASP centrality measures. Table 5 shows the ranking results for Email, Yeast, and Router networks, while the ranking results for Twitter, Rice-Facebook, and DBLP networks are given in Table 6. In the analysis of the Email network, SAASP shares 7 common nodes with BC, CC, and CI in the top-10 ranking. This similarity is observed with DC and SLRASP, showing 10 and 8 common nodes, respectively. SAASP reports an identical ranking compared to DC, while it shares the most differences with BC, CC, and CI. Meanwhile, when examining the top 10 nodes for

Table 4
Centrality scores of SAASP and SLRASP based on different neighborhood levels

Networks	SLRASP			SAASP		
	$L = 2$	$L = 3$	$L = 4$	$L = 2$	$L = 3$	$L = 4$
Email	0.6623 ± 0.04	0.6809 ± 0.04	0.6625 ± 0.02	0.7479 ± 0.03	0.7521 ± 0.02	0.7241 ± 0.04
Yeast	0.6647 ± 0.01	0.6677 ± 0.04	0.6624 ± 0.03	0.7311 ± 0.03	0.7242 ± 0.03	0.7185 ± 0.02
Router	0.6354 ± 0.03	0.6570 ± 0.03	0.6080 ± 0.01	0.6837 ± 0.03	0.7362 ± 0.01	0.7373 ± 0.02
Twitter	0.5884 ± 0.03	0.6302 ± 0.03	0.5721 ± 0.02	0.6271 ± 0.01	0.6513 ± 0.02	0.6560 ± 0.04
Rice-Facebook	0.6435 ± 0.04	0.6502 ± 0.04	0.6060 ± 0.02	0.6932 ± 0.02	0.7010 ± 0.03	0.7003 ± 0.03
DBLP	0.571 ± 0.02	0.5523 ± 0.04	0.5935 ± 0.03	0.5806 ± 0.02	0.5601 ± 0.04	0.5757 ± 0.02
Average	0.6275 ± 0.04	0.6397 ± 0.04	0.6174 ± 0.02	0.6773 ± 0.04	0.6875 ± 0.01	0.6853 ± 0.05

Table 5
Top-10 ranking nodes for email, yeast, and router networks

Rank	Email network						Yeast network						Router network					
	DC	BC	CC	CI	SLRASP	SAASP	DC	BC	CC	CI	SLRASP	SAASP	DC	BC	CC	CI	SLRASP	SAASP
1	105	333	333	105	105	105	20	447	216	597	597	597	101	3	3	101	90	101
2	333	105	23	333	333	333	249	21	36	249	249	249	140	1	101	140	385	90
3	16	23	105	23	23	23	442	36	616	582	582	582	351	101	90	3	351	385
4	23	578	42	42	42	42	21	20	249	21	428	36	63	140	140	90	357	351
5	42	76	41	41	299	16	36	442	21	20	86	21	49	160	1	1	63	140
6	41	233	76	76	76	76	447	216	20	442	280	428	243	509	243	243	280	1
7	196	135	233	233	21	233	97	249	582	36	83	20	114	100	385	100	382	136
8	233	41	52	52	233	41	428	104	442	616	436	436	136	351	427	63	3	49
9	21	355	135	135	52	196	582	309	814	428	251	251	1	63	100	385	368	3
10	76	42	378	378	135	21	86	428	86	1	570	86	52	180	217	351	370	357

Table 6
Top-10 ranking nodes for Twitter, Rice-Facebook, and DBLP networks

Rank	Twitter network						Rice-Facebook network						DBLP network					
	DC	BC	CC	CI	SLRASP	SAASP	DC	BC	CC	CI	SLRASP	SAASP	DC	BC	CC	CI	SLRASP	SAASP
1	9	63	63	504	504	504	124	313	313	127	313	124	39	2301	2301	234	3471	3471
2	18	9	9	63	63	63	127	547	607	124	127	127	78	3159	3159	78	273	273
3	27	18	18	54	54	54	313	979	547	130	547	313	117	3471	3471	156	39	39
4	63	54	27	126	126	126	130	607	247	154	124	607	156	624	4173	39	624	624
5	54	72	54	27	9	9	531	355	973	157	607	547	234	663	273	273	4095	4095
6	72	99	72	9	189	27	892	127	550	160	247	247	273	975	39	4095	156	156
7	162	126	99	189	72	189	893	247	127	163	550	130	312	273	624	624	78	78
8	189	153	126	18	90	18	894	973	133	199	973	550	624	39	663	117	2106	2106
9	99	189	90	90	18	72	355	133	640	202	979	154	2106	3627	975	3471	117	117
10	90	27	153	72	27	90	424	910	655	607	130	973	2652	2652	3627	312	312	312

both SAASP and SLRASP, nodes 105, 333, 23, and 42 are the first four nodes in the ranking list for both methods. In the Yeast network, the top-3 nodes ranked by SAASP, CI, and SLRASP are the same, with nodes 597, 249, and 582 identified as the key seed nodes in this network. In the Router network, SAASP and SLRASP have 7 common members in their

rankings, while the greatest difference for SAASP is observed with BC. Similarly, in the Twitter network, SAASP, DC, BC, and CC share 8 of the top-10 nodes. Also, the top-10 node list for CI, SLRASP, and SAASP is exactly the same, the only difference being in their order. The results also show that 9 common nodes appear in the top-10 list for the Rice-

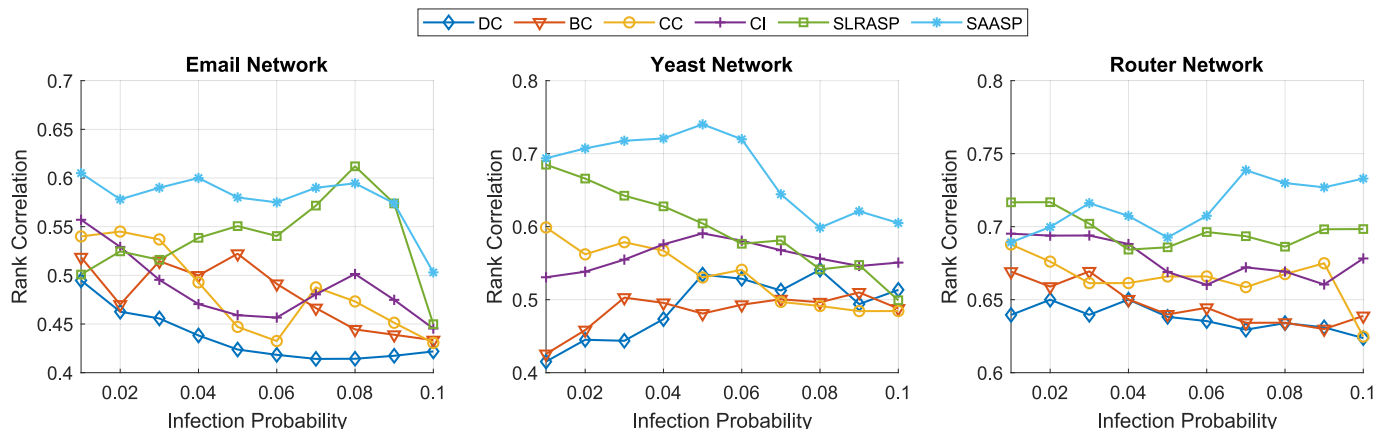


Fig. 5. Correlation between SAASP and other centrality measures over Email, Yeast, and Router networks

Facebook network between SAASP and SLRASP. Notably, SAASP and CI share the same 5 nodes in their top rankings for this network. Furthermore, in the DBLP network, SAASP shares 8, 4, 4, 9, and 10 nodes in the top-10 with DC, BC, CC, CI, and SLRASP, respectively.

As stated in [2], the simulations assume a recovery rate (μ) of 1, while the fixed infection rate (λ) is varied within the range of 0.01 to 0.1. The results of these simulations are presented in Fig. 5, which highlights the performance of the Kendall's τ coefficient when comparing the rankings produced by SAASP and other centrality measures, including DC, BC, CC, CI, and SLRASP. These evaluations were conducted across three different network datasets: Email, Yeast, and Router. This figure provides a detailed comparison of how closely the centrality rankings from each method align with the rankings generated by the SIR epidemic model. The same results for Twitter, Rice-Facebook, and DBLP networks are presented in Fig. 6. This simulation investigates the relationship between the influence measured by the function $F(t)$ and various centrality metrics by analyzing the rank correlation at different λ probabilities. By varying the value of λ , the simulation evaluates how closely the influence rankings, derived from $F(t)$ in the epidemic model, align with those generated by traditional centrality measures. This correlation analysis at different λ values provide insights into the consistency of node rankings across different epidemic dynamics and centrality approaches, helping to better understand the robustness and accuracy of the SAASP metric in comparison to other methods.

The simulation results demonstrate that the proposed SAASP centrality measure has little to no correlation with traditional centrality metrics such as DC, BC, CC, and CI, indicating that it captures unique aspects of network structure. This lack of correlation highlights its distinct approach to identifying influential nodes, especially in large-scale networks. SAASP's efficiency and superiority over conventional measures can be attributed to its innovative methodology, which combines both local and semi-local information. This is achieved through the integration of an augmented graph representation, the distributed extraction of local subgraphs, and the application of ASP theory, all of which contribute to more accurate and comprehensive node rankings in complex networks. These advanced features enable SAASP to better access node influence in networks where traditional methods may fall short, particularly when dealing with large-scale and distributed systems. Meanwhile, the superiority of SAASP over advanced centrality measures like SLRASP in the simulations can be explained by several critical aspects. One of the primary reasons is SAASP's ability to utilize augmented graphs, allowing it to account for global relationships that extend beyond immediate local neighborhoods. This enhanced representation captures connections between distant nodes, integrating both structural and semantic similarities. Unlike SLRASP, which primarily focuses on more simplified local substructures and does not fully consider the broader network dynamics, SAASP's approach provides a

more comprehensive perspective on node influence. By factoring in these global relationships, SAASP offers a more nuanced and effective ranking of nodes, especially in networks with complex interactions where purely local measures like those used in SLRASP may fail to capture the full scope of node significance. On average, SAASP achieved a noticeable improvement in Kendall's τ coefficient when compared to traditional and advanced centrality measures. Specifically, it improved by 12.6 % over DC, 8.7 % over BC, 9.3 % over CC, 6.5 % over CI, and 1.7 % over SLRASP.

The simulation results for cumulative infected nodes, denoted as $F(t)$, over time for the Email, Yeast, and Router networks are depicted in Fig. 7, while those for the Twitter, Rice-Facebook, and DBLP networks are presented in Fig. 8. In all cases, the number of infected nodes increases over time and eventually reaches a steady state. The top 10 most influential nodes, identified by each centrality measure, are selected as seed nodes to observe the convergence of node propagation capabilities. SAASP demonstrates highly efficient propagation across most of the network and requires fewer steps to reach the steady state compared to other centrality measures. Furthermore, the results indicate that the proposed SAASP method consistently outperforms SLRASP across all networks in terms of propagation efficiency.

In another experiment, the runtime of various centrality measures on six real-world networks is compared to evaluate their computational complexity. The results of this comparison are presented in Table 7. The findings indicate that only local centrality measure (i.e., DC) has a lower runtime compared to other centrality measures because DC operates based on local network information. In calculating DC, it is sufficient to count the number of edges connected to each node, which only requires examining the direct neighbors of a node. This simplicity makes the computations faster as it relies solely on data from the node and its adjacent edges. In contrast, global centrality measures like BC, CC, and CI require more extensive information from the entire network or larger portions of it. These measures assess the influence of a node by considering indirect connections and paths between nodes, which makes the computations more complex and time-consuming. Therefore, it is obvious that DC has a lower runtime compared to BC, CC, and CI due to simpler calculations and limited reliance on broader network data. On the other hand, SAASP and SLRASP, which are classified as semi-local centrality measures, exhibit higher runtimes compared to DC but lower than BC, CC, and CI. This is because semi-local centrality measures, like SAASP and SLRASP, strike a balance between using both local and some broader network information, resulting in moderate computational complexity. While they rely on more extensive network data than DC, they do not require the full global network analysis that BC, CC, and CI do, making them more efficient in comparison to fully global measures. Meanwhile, SAASP shows a shorter runtime compared to SLRASP due to the inclusion of modules such as distributed local

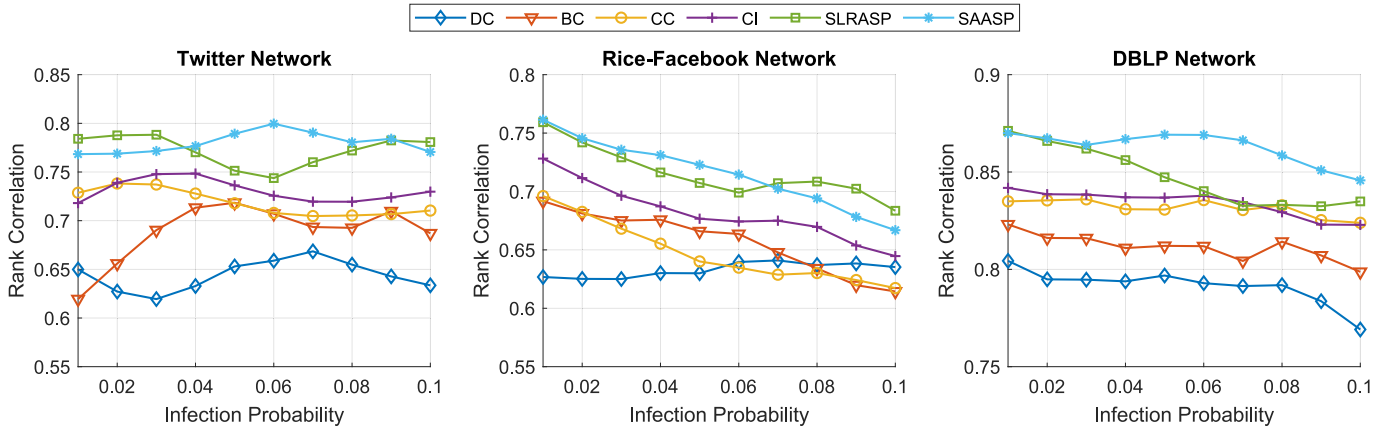


Fig. 6. Correlation between SAASP and other centrality measures over Twitter, Rice-Facebook, and DBLP networks

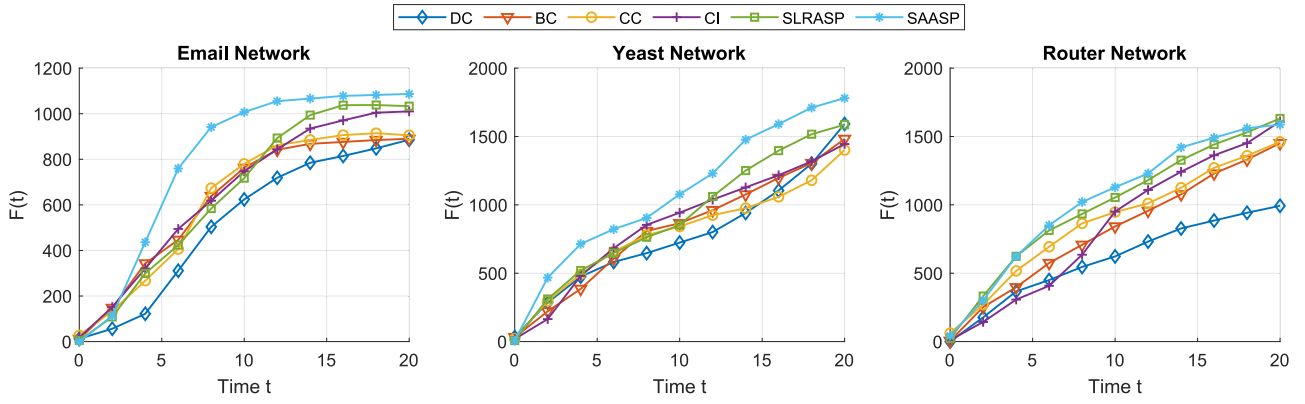


Fig. 7. Propagation influence of top-10 nodes as a function of time over Email, Yeast, and Router networks

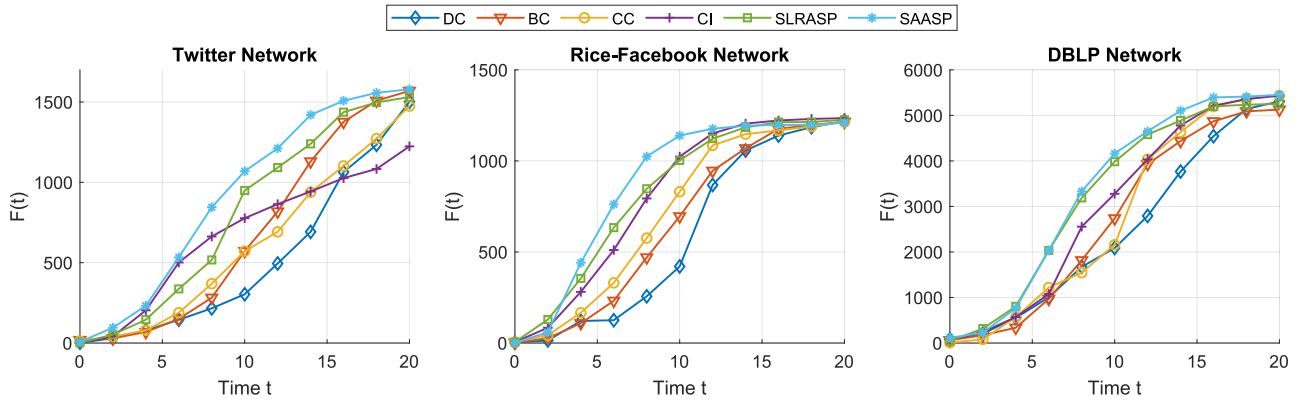


Fig. 8. Propagation influence of top-10 nodes as a function of time over Twitter, Rice-Facebook, and DBLP networks

Table 7
Running time results for different centrality measures

Dataset	DC	BC	CC	CI	SLRASP	SAASP
Email	0.81	9.15	9.96	11.67	6.91	6.16
Yeast	2.08	35.76	9.68	13.05	9.29	7.92
Router	1.82	30.96	9.09	11.31	8.02	6.67
Twitter	2.01	26.49	7.88	10.62	6.2	6.22
Rice-Facebook	2.39	20.22	7.55	10.38	6.05	5.9
DBLP	15.33	180.63	35.55	40.68	27.93	23.14
Average	4.07	50.54	13.29	16.29	10.73	9.34

subgraph extraction and semi-local ASP theory. Although SLRASP has a runtime comparable to SAASP, it does not always deliver the best performance in terms of influence spread, as it relies on a simplified approximation that does not fully capture the dynamics of influence propagation in large-scale and sparse networks. Moreover, SLRASP only considers structural features to identify influential nodes and does not account for semantic relationships.

5. Conclusions

In this paper, we addressed the critical problem of identifying influential nodes in complex networks, a task with significant theoretical and practical implications across various domains such as social networks, bioinformatics, and information dissemination. To tackle the limitations of existing centrality measures, we proposed SAASP (Semi-local Augmented Shortest Path), a novel centrality measure that combines an augmented graph with ASP theory. The augmented graph allows for the consideration of global semantic relationships, while the semi-local ASP structure ensures efficient and accurate identification

of influential nodes. Our experimental results, validated using the SI model and Kendall's τ coefficient on real-world datasets, demonstrate that SAASP outperforms existing centrality measures in terms of accuracy and computational efficiency, making it a superior choice for network analysis. Despite its advantages, the SAASP measure has certain challenges and limitations. One limitation is the potential computational overhead associated with constructing the augmented graph for extremely large-scale networks. Additionally, while SAASP effectively integrates global and local information, it may still face difficulties in dynamically changing networks where node relationships evolve over time. Another challenge lies in the precise tuning of parameters for different types of networks, which may require extensive experimentation. These limitations suggest several avenues for future research, including optimizing the augmented graph construction process, extending the SAASP measure to handle dynamic networks, and developing adaptive parameter tuning methods. Addressing these challenges will further enhance the applicability and robustness of the SAASP measure in identifying influential nodes in complex networks.

CRedit authorship contribution statement

Bo Meng: Data curation, Methodology, Resources, Validation, Writing – original draft, Writing – review & editing. **Amin Rezaeipannah:** Writing – review & editing, Validation, Methodology, Data curation.

Ethics approval

Not applicable.

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Declaration of competing interest

There is no free code for this study.

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Data availability

Not applicable.

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